Tut-5

Microprocessor Programming and Interfacing

Q1. Consider the following instructions. What will be the values of registers (AX, CX, DX, SP) after the following instructions are executed.
Assume following register values,
CS: 1500H, DS: 3200H, ES: 5000H, SS:8500H, IP: 1000H, AX: 1000H, BX:2000H, CX: 2000H, DX: 4000H, SI: 5000H, DI: 6000H, SP: 7000H, BP: 8000H

```
PUSH AX
PUSH DX
PUSHA
PUSH BX
POP DX
POPA
POP CX
```

Consider the following instructions. What will be the values of registers (AX, CX, DX, SP) after the following instructions are executed.
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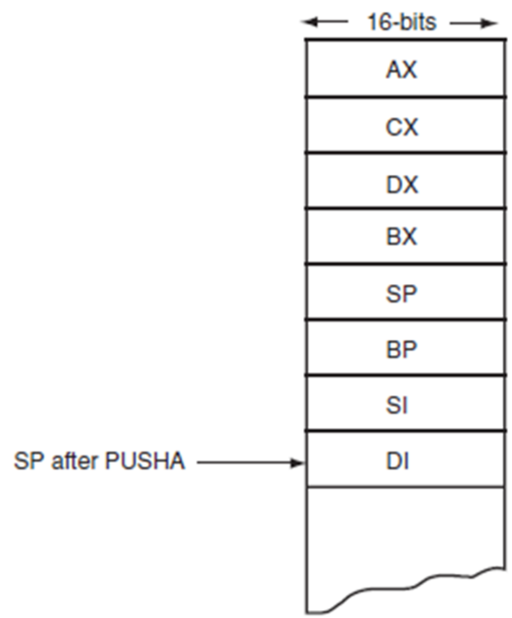
CS: 1500H, DS: 3200H, ES: 5000H, SS:8500H, IP: 1000H, AX: 1000H,
BX:2000H, CX: 2000H, DX: 4000H, SI: 5000H, DI: 6000H, SP: 7000H, BP: 8000H

```
PUSH AX
PUSH DX
PUSHA
PUSH BX
POP DX
POPA
POP CX
```



SS:6FFE → 1000H
SS:6FFC → 4000H
SS:6FFA → 1000H
SS:6FF8 → 2000H
SS:6FF6 → 4000H
SS:6FF4 → 2000H
SS:6FF2 → 6FFCH
SS:6FF0 → 8000H
SS:6FEE → 5000H
SS:6FEC → 6000H
SS:6FEA → 2000H ← SP

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Assume following register values,
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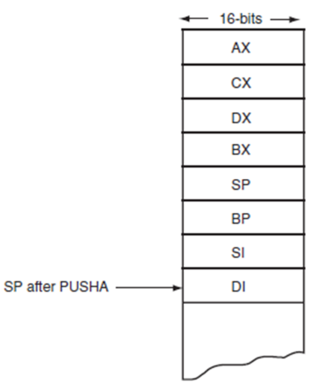
BX:2000H, CX: 2000H, DX: 4000H, SI: 5000H, DI: 6000H, SP: 7000H, BP: 8000H

```
POP DX
POPA
POP CX
```

After POP DXDX ← 2000H
SP = 6FEC



SS:6FFE → 1000H
SS:6FFC → 4000H
SS:6FFA → 1000H
SS:6FF8 → 2000H
SS:6FF6 → 4000H
SS:6FF4 → 2000H
SS:6FF2 → 6FFCH
SS:6FF0 → 8000H
SS:6FEE → 5000H
SS:6FEC → 6000H
SS:6FEA → 2000H ← SP

- DI ← 6000H
- SI ← 5000H
- BP ← 8000H
- (6FFCH popped but NOT loaded into SP)
- BX ← 2000H
- DX ← 4000H
- CX ← 2000H
- AX ← 1000H

SP becomes = 6FFCH

After POP CX
CX ← 4000H
SP = 6FFEH

Q2 Assume following register values,
CS: 1500H, DS: 4000H, ES: 5000H, SS:8500H, IP: 1000H, AX: 1000H, BX:2000H, CX: 2000H, DX: 4000H, SI: 3000H, DI: 6000H, SP: 7000H, BP: 8000H

Assume 12H is present in all memory locations from 40000H to 5FFFFH and 13H is present in all memory locations from 50000H to 6FFFFH before the following instructions are executed What memory locations/ register values (including SP) were changed by that. Please do note that following instructions are in sequence and happen one after another

Physical memory locations/ registers modified by the instruction and updated value

PUSH DWRD PTR[SI]
PUSH CX
POP DX
PUSH WRD PTR [BX]
POP DX

Q.3 Assume following register values,

CS: 1500H, DS: 3200H, ES: 5000H, SS:8500H, IP: 1000H, AX: 1000H, BX:2000H, CX: 3000H, DX: 4000H, SI: 5000H, DI: 6000H, SP: 7000H, BP: 8000H

Now consider the following instructions.

Mention the physical memory addresses/ registers that are modified by these instructions. Please do note that these instructions are subsequent (i.e. MOVSB is executed after INC WORDPTR [SI] and so on, or in other words they are not independent instructions but are part of an ALP)

Physical memory addresses/ registers modified by the instruction and their updated value

```
CLD
INC WORDPTR [SI]
MOVSB
STOSB
LODSW
SCASB
CMPSB
```

Q.3 Assume following register values,

CS: 1500H, DS: 3200H, ES: 5000H, SS:8500H, IP: 1000H, AX: 1000H, BX:2000H, CX: 3000H, DX: 4000H, SI: 5000H, DI: 6000H, SP: 7000H, BP: 8000H

Now consider the following instructions.

Mention the physical memory addresses/ registers that are modified by these instructions. Please do note that these instructions are subsequent (i.e. MOVSB is executed after INC WORDPTR [SI] and so on, or in other words they are not independent instructions but are part of an ALP) [8 Marks]

Physical memory addresses/ registers modified by the instruction and their updated value

CLD

INC WORDPTR [SI]

MOVSB

STOSB

LODSW

SCASB

CMPSB

1) CLD

DF = 0, No memory/register modification.

2) INC WORD PTR [SI]

Address = DS:SI
= 32000H + 5000H
= 37000H

Modified:
Memory at 37000H–37001H (word incremented by 1)

SI unchanged.

Q.3 Assume following register values,

CS: 1500H, DS: 3200H, ES: 5000H, SS:8500H, IP: 1000H, AX: 1000H, BX:2000H, CX: 3000H, DX: 4000H, SI: 5000H, DI: 6000H, SP: 7000H, BP: 8000H

Now consider the following instructions.

Mention the physical memory addresses/ registers that are modified by these instructions. Please do note that these instructions are subsequent (i.e. MOVSB is executed after INC WORDPTR [SI] and so on, or in other words they are not independent instructions but are part of an ALP) [8 Marks]

Physical memory addresses/ registers modified by the instruction and their updated value

```
CLD
INC WORDPTR [SI]
MOVSB
STOSB
LODSW
SCASB
CMPSB
```

3) MOVSB

Source: DS:SI = 32000H + 5000H = 37000H

Destination: ES:DI = 50000H + 6000H = 56000H

Modified:Memory 56000H ← Memory 37000H

SI = 5001H, DI = 6001H

4) STOSB

ES:DI = 50000H + 6001H = 56001H

Modified: Memory 56001H ← AL

DI = 6002H

Q.3 Assume following register values,

CS: 1500H, DS: 3200H, ES: 5000H, SS:8500H, IP: 1000H, AX: 1000H, BX:2000H, CX: 3000H, DX: 4000H, SI: 5000H, DI: 6000H, SP: 7000H, BP: 8000H

Now consider the following instructions.

Mention the physical memory addresses/ registers that are modified by these instructions. Please do note that these instructions are subsequent (i.e. MOVSB is executed after INC WORDPTR [SI] and so on, or in other words they are not independent instructions but are part of an ALP) [8 Marks]

Physical memory addresses/ registers modified by the instruction and their updated value

```
CLD
INC WORDPTR [SI]
MOVSB
STOSB
LODSW
SCASB
CMPSB
```

5) LODSW

DS:SI = 32000H + 5001H = 37001H

Modified:

AX ← word at 37001H–37002H

SI = 5003H

6) SCASB

ES:DI = 50000H + 6002H = 56002H

Modified:

Flags affected

DI = 6003H

Q.3 Assume following register values,

CS: 1500H, DS: 3200H, ES: 5000H, SS:8500H, IP: 1000H, AX: 1000H, BX:2000H, CX: 3000H, DX: 4000H, SI: 5000H, DI: 6000H, SP: 7000H, BP: 8000H

Now consider the following instructions.

Mention the physical memory addresses/ registers that are modified by these instructions. Please do note that these instructions are subsequent (i.e. MOVSB is executed after INC WORDPTR [SI] and so on, or in other words they are not independent instructions but are part of an ALP)

Physical memory addresses/ registers modified by the instruction and their updated value

```
CLD
INC WORDPTR [SI]
MOVSB
STOSB
LODSW
SCASB
CMPSB
```

7) CMPSB

Source:

DS:SI = 32000H + 5003H = 37003H

Destination:

ES:DI = 50000H + 6003H = 56003H

Modified:

Flags affected

SI = 5004H

DI = 6004H

Q4. An interleaved string is stored from displacement ‘istr1’. The size of the interleaved string is stored in location ‘cnt1’. Write an ALP that will separate the interleaved string into two strings as shown below. If the interleaved string is “hmeilclroo” it should be separated as two strings “hello” and “micro”. You can assume that the strings to be separated will be of equal size.

```
.model tiny
.data
instr1 db 'H','m','e','i','l','c','l','r','o','o'
cnt1 db 10
str1 db 5 dup(0)
str2 db 5 dup(0)

.code
.startup
lea si,instr1
lea di, str1
lea bx,str2
mov cl, cnt1
l1:
mov al,[si]
mov [di],al
inc si
mov al,[si]
mov [bx],al
inc si
inc bx
inc di
sub cl,2
jnz l1
.exit
end
```